Analysis of gender stereotypes in personality assessments of recorded speech

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Agenda

- Project motivation & aims
- The corpus
- Process
  - Statistical Analysis
  - Machine Learning
  - Within gender assessment
- Results & Conclusions
- Future work

Motivation/Aims

- Personality Perception in social psychology
- Analysis of Gender stereotypes in personality perception
  - Under zero acquaintace
  - Using non-verbal speech

The corpus *

- 640 audio recordings from a Swiss radio channel
- 11 non french speaking judges
- The Big Five personality triats**
  - Openness, Concienciousness, Extraversion, Agreeableness, Neuroticism
  - Five scores for every speaker from each judge

Sources:

* G. Mohammadi and A. Vinciarelli. IEEE transactions on affective computing 1 automatic personality perception: Prediction of trait attribution based on prosodic features. 2012.
** R. R. Mccrae and O. P. John. An introduction to the five-factor model and its applications.1992.

Process

- Project Aim:
  - **Analysis of gender stereotype influence on personality percpetion under zero acquiantance using the corpus**
- Process:
  - Statistical analysis
  - Machine Learning methodologies
  - Within gender assessment

Statistical Analysis

- Inter-rater agreement
  - Agreement between the 11 judges
  - Spearman's rank correlation & Spearman-Brown reliability
- Gender differences
  - Student's t tests
  - Imbalanced gender distribution in the dataset



Figure: A series of Box plots illustrating the average personality assessment scores given to audio recordings by the judges. Each plot represents a Big Five personality trait under which the scores were assigned which are separated into the two boxes dedicated to the binary genders. "" indicators suggest statistical significance according to the t-test results between the genders per trait.*

Machine Learning Approach

- Used the scikit-learn python library
- Five classification algorithms were used
  - Features: Big five scores
  - Labels: Binary gender
- Stratified group k-fold cross validation was used
- Hyperparameter tuning was done
- Evalutation of the model performance
  - Accuracy score & Binomial test
  - Student’s t-tests for all performance metrics

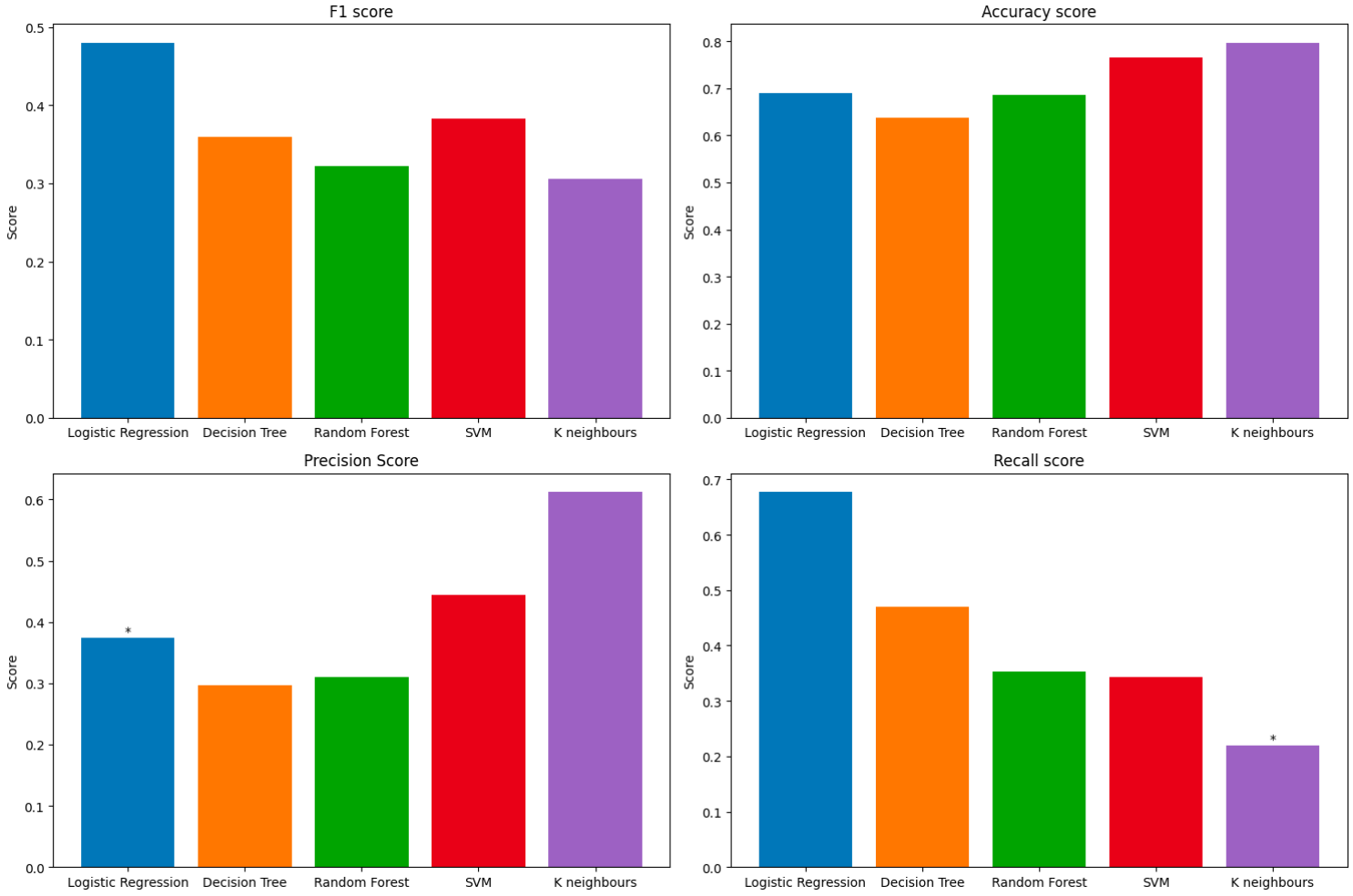


Figure: Bar charts for every performance metric for each classification algorithm's performance post hyperparameter tuning. The "*" indicators suggest statistical significance as a result of Student's t tests between the metric over the k-folds and the prior probability of 66%.

Within gender assessment

- Fundamental frequency & gender
- Within gender personality perception
- Within gender inter-rater agreement
- Fundemental frequency extraction using the openSmile* tool
- Spearman's rank to compare the Big five scores and fundamental frequency

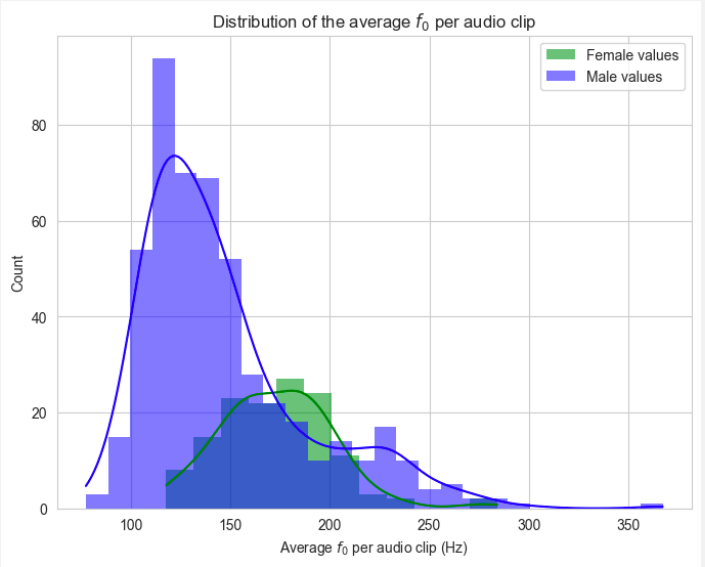


Figure: Distribution of average fundamental frequency ƒ0 values per audio recording for the two gender groups.

Sources:

* F. Eyben, M. Wöllmer, and B. Schuller. Opensmile: the munich versatile and fast open-source audio feature extractor. MM '10, page 1459–1462, New York, NY, USA, 2010. Association for Computing Machinery.

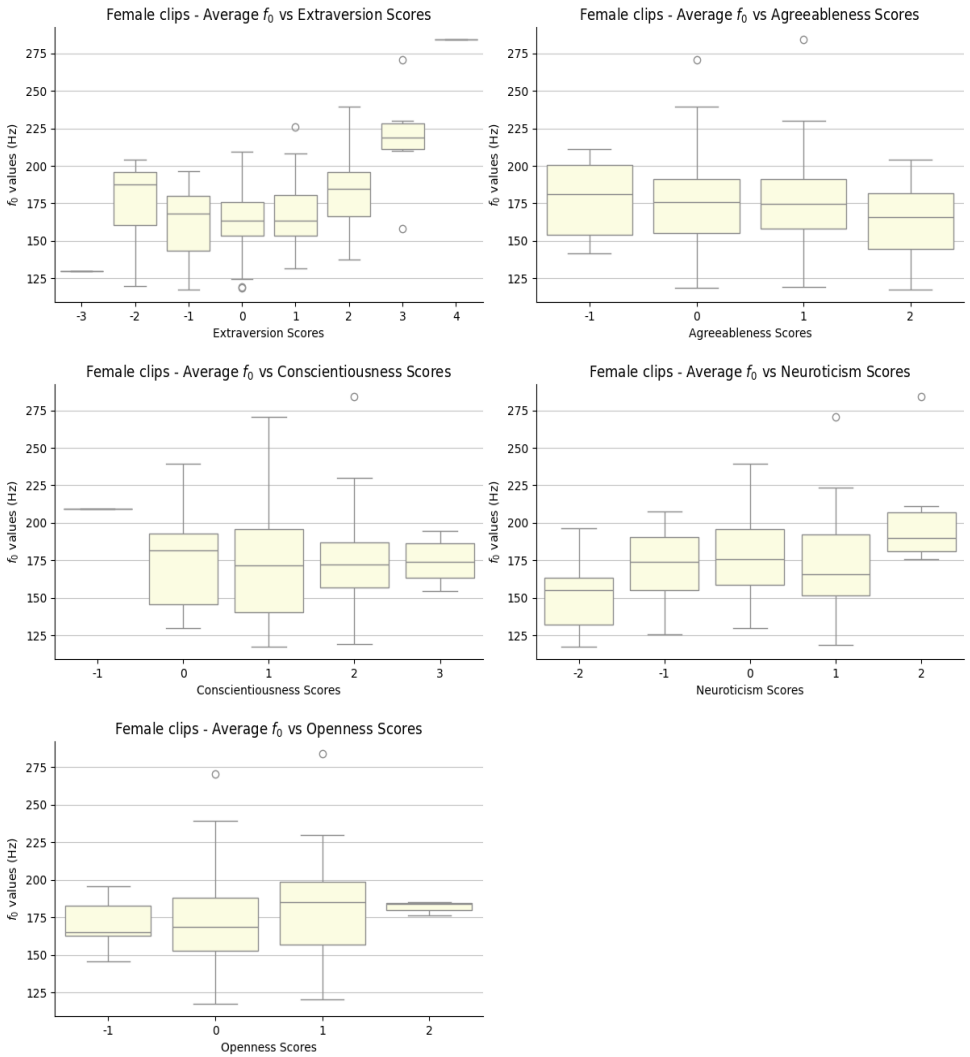


Figure: A series of Box plots illustrating the range of average ƒ0 values corresponding to each personality perception score for a specific trait assigned to female speakers.

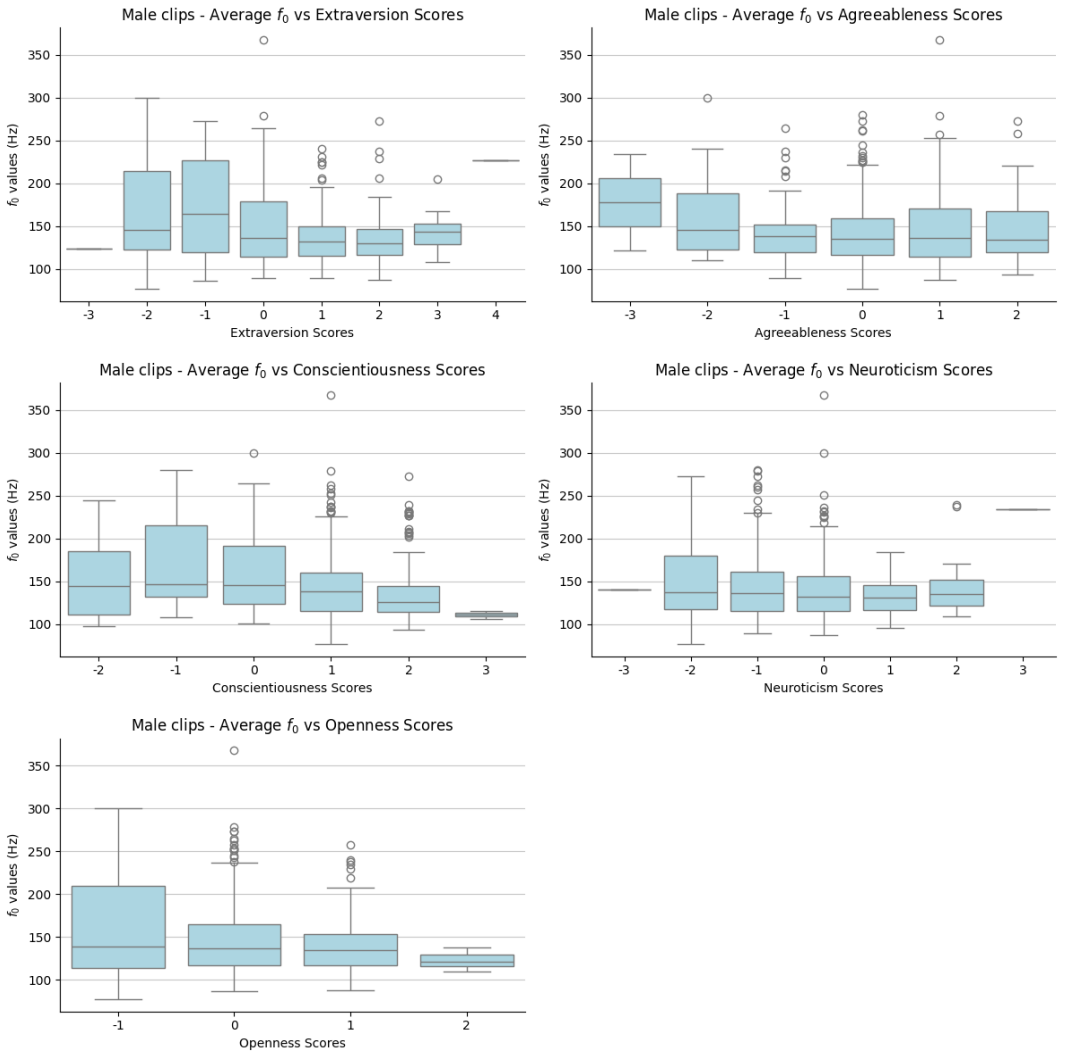


Figure: A series of Box plots illustrating the range of average ƒ0 values corresponding to each personality perception score for a specific trait assigned to male speakers.

The results & conclusions

- Gender stereotypes appear to influence personality perception under zero acquaintance
- The impact of gender stereotypes on within gender personality perception remains unclear

Future work

- Use of more extensive datasets to establish more conclusive results for the within gender impact of gender bias.
- Introduction of more inclusive and representative datasets of the gender spectrum
- Further investigation on the adoption of machine learning for the detection of gender stereotypes in datasets.

Thank you